

I-Logical Approaches Circuits

Foundations of Neuro-Symbolic AI

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt (THWS)

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Outline

- ▶ Circuit computation model
- ▶ Tensor representation
- ▶ Example: m -adic sums

Propositional Syntax

Propositional syntax decomposes formulas into connectives applied on sub-formulas.

When understanding formulas as functions, we get a decomposition into circuits of connectives.

This decomposition is reflected in the tensor network decomposition of basis encodings.

Can we generalize to arbitrary functions and find tensor network decompositions based on circuits?

Example: Neural networks are circuits of neurons.

Circuits as a computational model

Circuits are a fundamental computational model, which is used in many areas of computer science, e.g., in complexity theory, logic, and machine learning.

Other models:

- ▶ Turing machines

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- ▶ Example: m -adic sums

Set enumerating variables

Generic basis encodings of functions contain set enumerating variables to the input and target sets.

Computation hypergraph

- ▶ Nodes represent inputs, preliminary results, and outputs
- ▶ Edges are directed and represent

Formally: A circuit is a decorated hypergraph.

Decomposition into tensor networks

The basis encoding to a function of a circuit is decomposed into basis encodings of the node functions.

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- ▶ Example: m -adic sums

m -adic representation of integers

We develop a tensor network representation of integer summations on the set $[m^d] = \{0, \dots, m^d - 1\}$, where $m, d \in \mathbb{N}$,

$$+ : [m^d] \times [m^d] \rightarrow [m^{d+1}] \quad , \quad +(i, \tilde{i}) = i + \tilde{i},$$

which have a m -adic representation of length d . We define an index interpretation map

$$I : \bigtimes_{k \in [d]} [m] \rightarrow [m^d] \quad , \quad I(x_{[d]}) = \sum_{k \in [d]} x_k \cdot m^k,$$

which enables the parameterization of $[m^d]$ as the states of d categorical variables $X_{[d]}$ of dimension m .

Basis encoding

The basis encoding of the sum is then

$$\begin{aligned} & \beta^+ [Y_{[d+1]}, X_{[d]}, Y_{[d]}] \\ &= \sum_{x_{[d]}, \tilde{x}_{[d]}} \epsilon_{l^{-1}(l(x_{[d]})+l(\tilde{x}_{[d]}))} [Y_{[d+1]}] \otimes \epsilon_{x_{[d]}} [X_{[d]}] \otimes \epsilon_{\tilde{x}_{[d]}} [\tilde{X}_{[d]}] . \end{aligned}$$

Decomposition into half-adders

The targeted function to be decomposed is the representation of the integer sum by

$$+^m : \left(\prod_{k \in [d]} [m] \right) \times \left(\prod_{k \in [d]} [m] \right) \rightarrow \prod_{k \in [d+1]} [m] \quad , \quad +^m(x_{[d]}, \tilde{x}_{[d]}) = I^{-1}(I(x_{[d]}, \tilde{x}_{[d]}))$$

We build a decomposition hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ (see Def. ??) consisting of $4 \cdot d$ nodes (see Figure ??a). The nodes carry the $(3 \cdot d + 1)$ variables $X_{[d]}, Y_{[d]}, Y_{[d+1]}$ of dimension m constructed above and $d - 1$ auxiliary variables $Z_{[d-1]}$ of dimension 2 representing carry bits. The directed hyperedges of $\mathcal{G}$ are

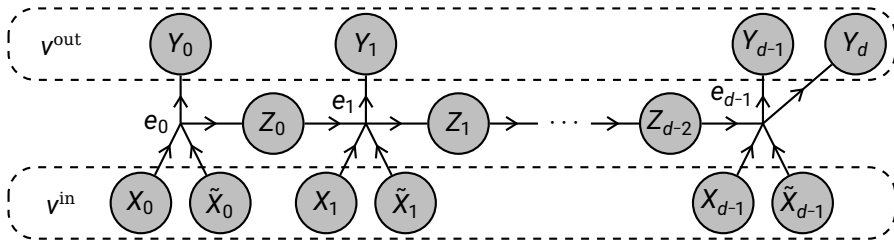
$$\begin{aligned} \mathcal{E} = & \left\{ (\{X_0, \tilde{X}_0\}, \{Y_0, Z_0\}) \right\} \cup \left\{ (\{Z_{k-1}, X_k, \tilde{X}_k\}, \{Y_k, Z_k\}) : k \in \{1, \dots, d-1\} \right\} \\ & \cup \left\{ (\{Z_{d-2}, X_{d-1}, \tilde{X}_{d-1}\}, \{Y_{d-1}, Y_d\}) \right\} \end{aligned}$$

and are decorated by local summation functions

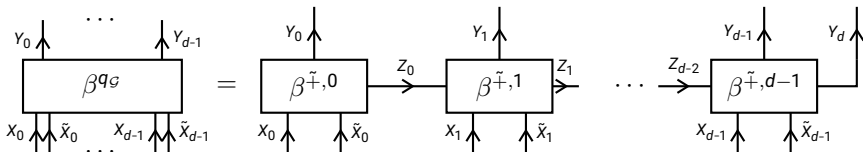
$$\tilde{+} : [2] \times [m] \times [m] \rightarrow [m] \times [2] \quad , \quad \tilde{+}(z, x, \tilde{x}) = \left((z + x + \tilde{x}) \bmod m, \right.$$

Tensor Network representation

a) Hypergraph representing the circuit:



b) Basis encoding with decomposition:



By Thm. ?? we have a decomposition of the basis encoding to q_G (see

Extension to multiplications

Do the multiplication for each digit of the right number separately, then add the results.

In this way: Generalize to